

MSE 220 Lab Report: Materials Science in Practice

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Class: MSE 220 D100

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Executive Summary

This report presents the comprehensive outcomes and insights garnered from participating in the MSE 220 lab series. Under the expert guidance of Dr. Ahad Armin and our committed team of teaching assistants, our group has delved deeply into the practical aspects of materials science, an essential discipline in engineering. Each session in this series was meticulously crafted to focus on a distinct element of materials science, encompassing a range from 3D printing methodologies to the intricacies of tensile and fracture toughness testing. This document not only records our collective efforts but also reflects our journey through these varied, yet interrelated, components of materials science.

Lab 05 provided a dual perspective on tensile properties through both experimental testing and simulation using Solidworks 2018. The objective was to measure the yield strength and ultimate strength of 3D printed specimens and compare these with simulated results. This lab enhanced our understanding of the tensile properties of materials, the accuracy of simulations, and the influence of factors like infill density on material behavior under stress.

In Lab 6, we delved into the fracture toughness of materials, focusing on steel samples with different notch shapes. The lab's primary goal was to understand how notches affect the stress and deformation characteristics of materials. The findings highlighted the variations in impact energy resistance between V-shaped and U-shaped notches, illustrating the critical role of notch geometry in material failure.

Throughout these labs, our team gained valuable hands-on experience and theoretical knowledge in materials science. Each lab session was a building block towards understanding the complex interactions between material properties, manufacturing processes, and their applications in engineering. The series not only enriched our academic knowledge but also provided practical skills and analytical abilities essential for future endeavors in the field of engineering and materials science.

This report stands as a testament to our learning journey in MSE 220, showcasing our efforts, discoveries, and the comprehensive knowledge gained in the realm of materials science.

Lab 05: Tensile Testing and Simulation

Introduction

MSE220 Lab 05 focused on tensile testing and its simulation using 3D printed dogbone samples. This lab session involved both the practical application of tensile testing on physical specimens and the simulation of tensile testing using Solidworks, providing a comprehensive understanding of the tensile properties of materials.

Objectives

- Fill the Table by Measuring Yield Strength and Ultimate Strength: Measure and record the yield strength and ultimate strength of the samples during the tensile tests.
- Discuss Differences or Coincidences Between Simulation and Experimental Results: Analyze and compare the outcomes of both simulation and experimental approaches, focusing on the reasons behind any observed discrepancies or similarities.

Methodology

Preparation of the Simulation Model for Tensile Test

- CAD Design: Utilized Solidworks to prepare CAD designs with inner designs of infill 20% and 40%. The design was made to match the actual printed sample.
- Simulation Settings: Configured initial simulation settings, boundary conditions, and external load settings in Solidworks.

Tensile Testing Experiment with 3D Printed Dogbone Samples

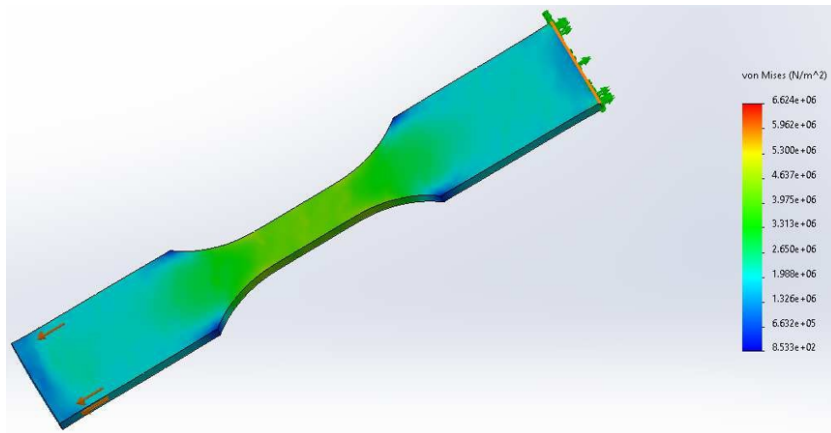
- Sample Preparation: CAD designs of dogbone samples were prepared with fully filled inner designs. These designs were exported as .stl files and printed with specified infill densities (20% and 40%).
- Material Selection: Various materials were used, with specific printing temperatures for each (e.g., PLA, ABS, PVA, TPU 95A).
- Conducting Tensile Test: The thickness, width, and length of samples were measured before testing. Samples were then fixed in a universal test machine, and tensile tests were conducted at a loading speed of less than 2 mm/sec.

Data Collection and Analysis

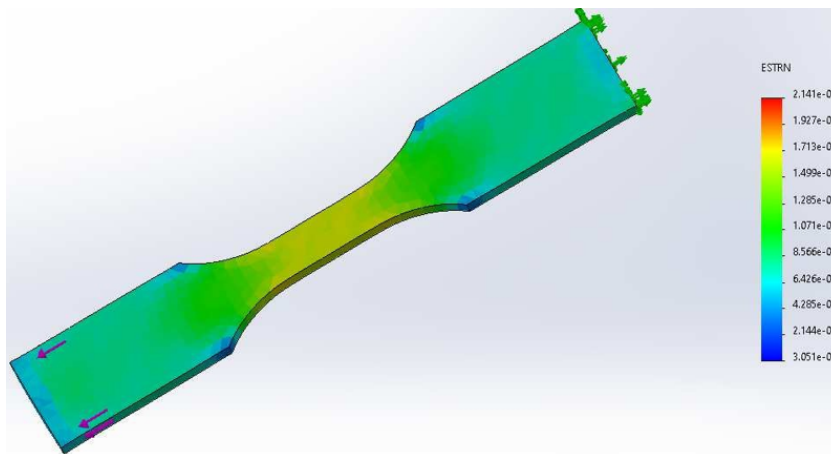
- Simulation Data: Collected simulation results from Solidworks, focusing on displacement (URES) and stress (von Mises) measurements.

20% Infill Simulated Tensile Test (50 N load):

Stress:

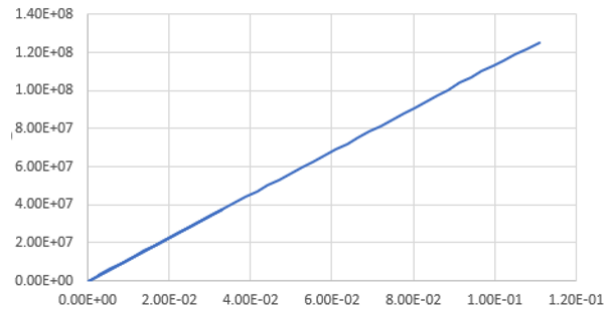


Strain:



Force	Stress	Strain	Modulus of Elasticity
25	3.14E+06	2.77E-03	1.13E+09
50	6.27E+06	5.54E-03	1.13E+09
75	9.41E+06	8.31E-03	1.13E+09
100	1.25E+07	1.11E-02	1.13E+09
125	1.57E+07	1.39E-02	1.13E+09
150	1.88E+07	1.66E-02	1.13E+09
175	2.20E+07	1.94E-02	1.13E+09
200	2.51E+07	2.22E-02	1.13E+09
225	2.82E+07	2.49E-02	1.13E+09
250	3.14E+07	2.77E-02	1.13E+09
275	3.45E+07	3.05E-02	1.13E+09
300	3.76E+07	3.32E-02	1.13E+09
325	4.08E+07	3.60E-02	1.13E+09
350	4.39E+07	3.88E-02	1.13E+09
375	4.70E+07	4.15E-02	1.13E+09
400	5.02E+07	4.43E-02	1.13E+09
425	5.33E+07	4.71E-02	1.13E+09
450	5.64E+07	4.98E-02	1.13E+09
475	5.96E+07	5.26E-02	1.13E+09
500	6.27E+07	5.54E-02	1.13E+09
525	6.58E+07	5.82E-02	1.13E+09
550	6.90E+07	6.09E-02	1.13E+09
575	7.21E+07	6.37E-02	1.13E+09
600	7.53E+07	6.65E-02	1.13E+09
625	7.84E+07	6.92E-02	1.13E+09
650	8.15E+07	7.20E-02	1.13E+09
675	8.47E+07	7.48E-02	1.13E+09
700	8.78E+07	7.76E-02	1.13E+09
725	9.09E+07	8.03E-02	1.13E+09
750	9.41E+07	8.31E-02	1.13E+09

Simulation Stress vs Strain Curve

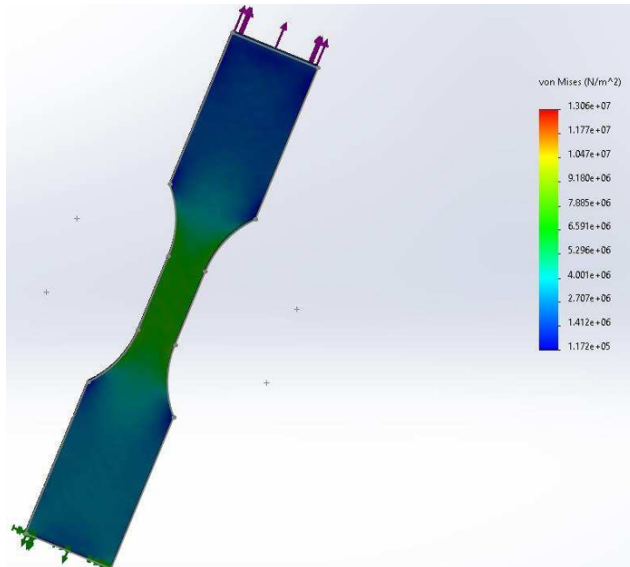


Modulus of Elasticity = Stress/ Strain = 1.13×10^9 Pa

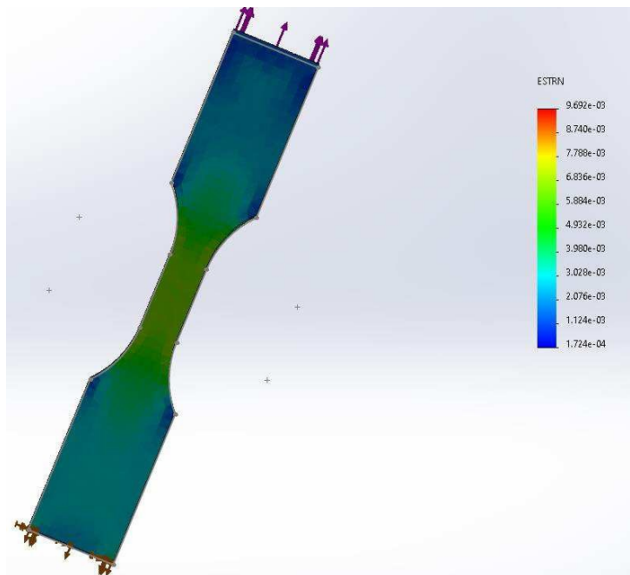
Ultimate Stress = 43.9 Mpa

40% Infill Simulated Tensile Test:

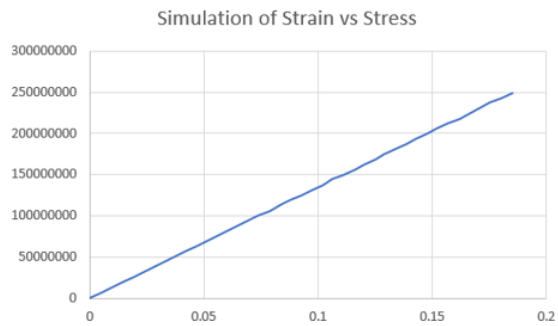
Stress:



Strain:



Force	Stress	Strain	Modulus of Elasticity
0	0	0	NA
25	6.78E+06	5.02E-03	1.35E+09
50	1.30E+07	9.60E-03	1.35E+09
75	1.92E+07	1.42E-02	1.35E+09
100	2.54E+07	1.88E-02	1.35E+09
125	3.17E+07	2.34E-02	1.35E+09
150	3.79E+07	2.81E-02	1.35E+09
175	4.41E+07	3.27E-02	1.35E+09
200	5.03E+07	3.73E-02	1.35E+09
225	5.65E+07	4.19E-02	1.35E+09
250	6.28E+07	4.65E-02	1.35E+09
275	6.90E+07	5.11E-02	1.35E+09
300	7.52E+07	5.57E-02	1.35E+09
325	8.14E+07	6.03E-02	1.35E+09
350	8.76E+07	6.49E-02	1.35E+09
375	9.38E+07	6.95E-02	1.35E+09
400	1.00E+08	7.41E-02	1.35E+09
425	1.06E+08	7.87E-02	1.35E+09
450	1.13E+08	8.33E-02	1.35E+09
475	1.19E+08	8.79E-02	1.35E+09
500	1.25E+08	9.25E-02	1.35E+09
525	1.31E+08	9.72E-02	1.35E+09
550	1.37E+08	1.02E-01	1.35E+09
575	1.44E+08	1.06E-01	1.35E+09
600	1.50E+08	1.11E-01	1.35E+09
625	1.56E+08	1.16E-01	1.35E+09
650	1.62E+08	1.20E-01	1.35E+09
675	1.68E+08	1.25E-01	1.35E+09
700	1.75E+08	1.29E-01	1.35E+09
725	1.81E+08	1.34E-01	1.35E+09
750	1.87E+08	1.39E-01	1.35E+09

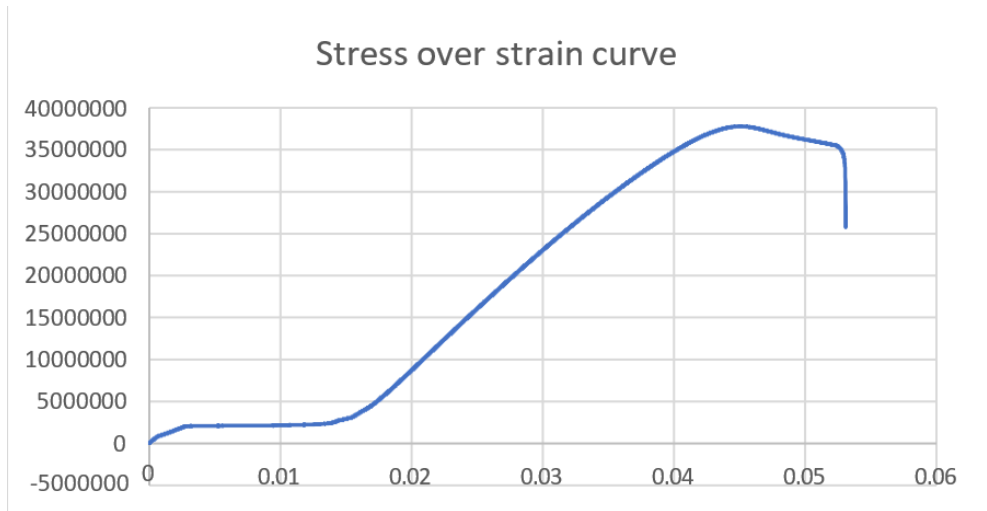


Modulus of Elasticity = Stress/ Strain = 1.35×10^9 Pa

Ultimate Stress = 62.8 Mpa

- Experimental Data: Recorded the force-displacement data during the tensile test and converted these to stress-strain curves for analysis.

20% Infill Tensile Test:

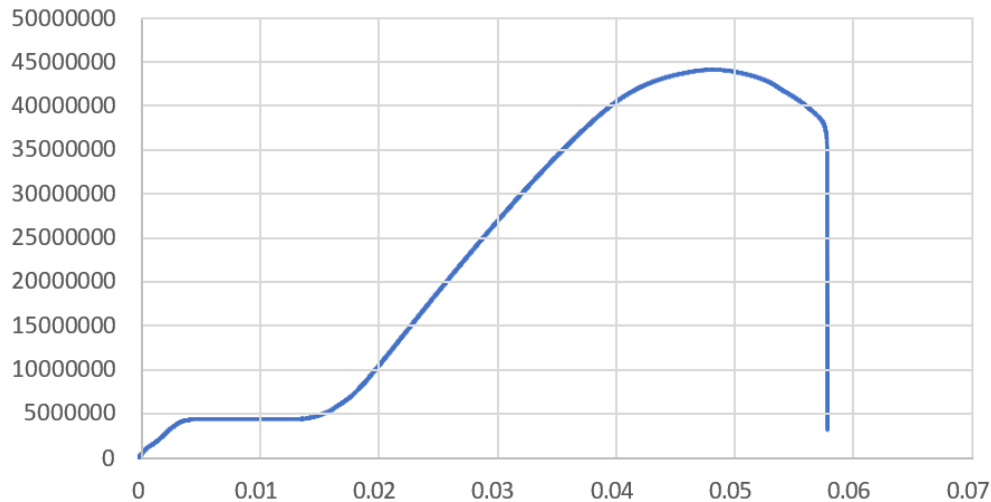


According to the data, the tensile strength of the part was **37.78 MPa**

Using the data provided we can plot the stress vs strain curve and use the values to calculate the modulus of elasticity, this being **1364.66 MPa**

40% Infill Tensile Test

Stress over Strain curve



According to the data, the tensile strength of the part was **44.17 MPa**

Using the data provided we can plot the stress vs strain curve and use the values to calculate the modulus of elasticity, this being **1511.32 MPa**

- Comparison and Discussion: The data from both simulations and experiments were compared to understand the behavior of materials under tensile stress.

Key Findings

1. Understanding Tensile Properties: Both the simulation and experimental approaches provided valuable insights into the tensile properties of the materials, including yield strength and ultimate strength.
2. Comparison of Simulation and Experimental Results: The comparison between the simulation and experimental results offered a deeper understanding of the material behavior and the accuracy of simulation in predicting real-world scenarios.
3. Effect of Infill Density: The lab demonstrated how variations in infill density can affect the tensile properties of 3D printed materials, thereby influencing their performance under stress.

Conclusion

Lab 05 in the MSE220 series provided a multifaceted understanding of tensile testing, encompassing both theoretical simulation and practical experimentation. Through the process of designing, simulating, and testing dogbone samples, students gained valuable insights into the mechanical behavior of materials under tensile loads. This lab not only reinforced the concepts of tensile properties but also highlighted the importance of simulation in materials science, offering a comprehensive view of how materials behave in tension and the factors that influence their performance.

Run	Infill (%)	Approach	Young's modulus	Ultimate Strength
1	20%	Simulation	1130MPa	43.9MPa
2	40%	Simulation	1350MPa	62.8MPa
3	20%	Experiment	1364.64MPa	37.78MPa
4	40%	Experiment	1511.3 MPa	44.17MPa

Lab 6 : Fracture Toughness Testing

Discussion of objectives described above using the provided experimental impact test data :

- As the result of observation at the steel sample with two different notch shapes, V-shape and U-shape, it is clear that the V-shaped notch can bear a greater force than the U-shaped one. The U-shaped notch fractured at an impact energy of 15 Nm, while the V-shaped notch fractured at an impact energy of 21 Nm. This discrepancy may be related to the U-shaped notch having two parallel sidewalls as well as a radius, resulting in a smaller cross-sectional area for the U-shaped specimen, making it more prone to failure. Despite this, both materials exhibited some bending, emphasizing the steel's ductility.

True and False Questions :

1. The concentration of stress in the notch root is substantially increased by inserting a notch into the specimen. “True” or “False”? **True**, because inserting a notch into a specimen often raises the stress concentration at the notch root. A notch generates a stress concentration point, which can result in greater localized stress than in the surrounding material.
2. The sharper the notch, the greater the stress in the notch root. “True” or “False”? **True**, A sharper notch causes a greater geometric disruption in the material, resulting in more stress at that particular spot.
3. The sharper notch produces a faster deformation speed in the notch root. “True” or “False”? **True** in terms of the Charpy test, The sharpness of a notch has little effect or not at all on the deformation speed near the notch root but, not for the Charpy testing. The general notch in a Charpy test is V & U model. The V model has more influence because of the concentrated area at the section.
4. The notched bar impact work of brass is twice as high as that of the steel specimen. “True” or “False”? **False**, the tensile strength of steel is more than that of brass.
5. All the brass specimens reveal brittle fractures at room temperature. “True” or “False”? **True**, brass specimens show brittle fracture because it reduces ductility at room temperature because of copper and zinc.
6. Tempering steel and structural steel tend to manifest ductile behavior at room temperature.

“True” or “False”? **True**, Tempered steel and structural steel are both normally designed to be ductile at room temperature.

7. Machining steel reveals a larger ductile fracture component at room temperature. “True” or “False”? **False**, Steel machining at room temperature does not always disclose a bigger ductile fracture component.
8. In summary, the different types of steel manifest primarily mixed fractures, although the brass specimens are brittle. “True” or “False”? **True**, Different varieties of steel can exhibit a variety of fracture behaviors, although they are often characterized by a combination of ductile and brittle characteristics, depending on factors such as steel type, heat treatment, and loading circumstances.